Cycle In Graph

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The problem

- Directed graph as adjacency list; does it contain a cycle?
  3-colour DFS.

Intuition

- A cycle exists if depth-first search meets a node that's still on the current recursion stack.

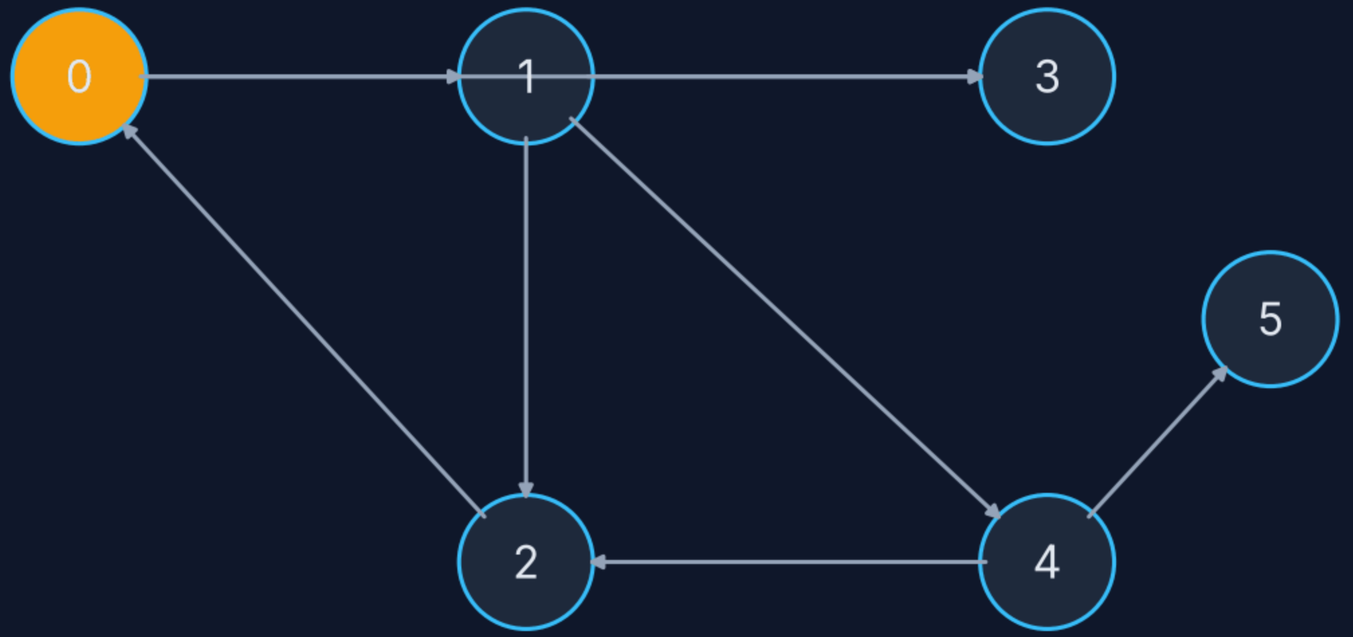
Brute force vs optimal

- Brute force: from each node, DFS to see if you return → $O(V \cdot (V+E))$
- Single DFS with three colours → $O(V + E)$
- Alternative: Kahn's topological sort fails ⇒ cycle

Approach

- 1. Colour nodes white, grey, black
- 2. Grey means on the stack
- 3. Edge to a grey node is a back edge, a cycle

Worked example



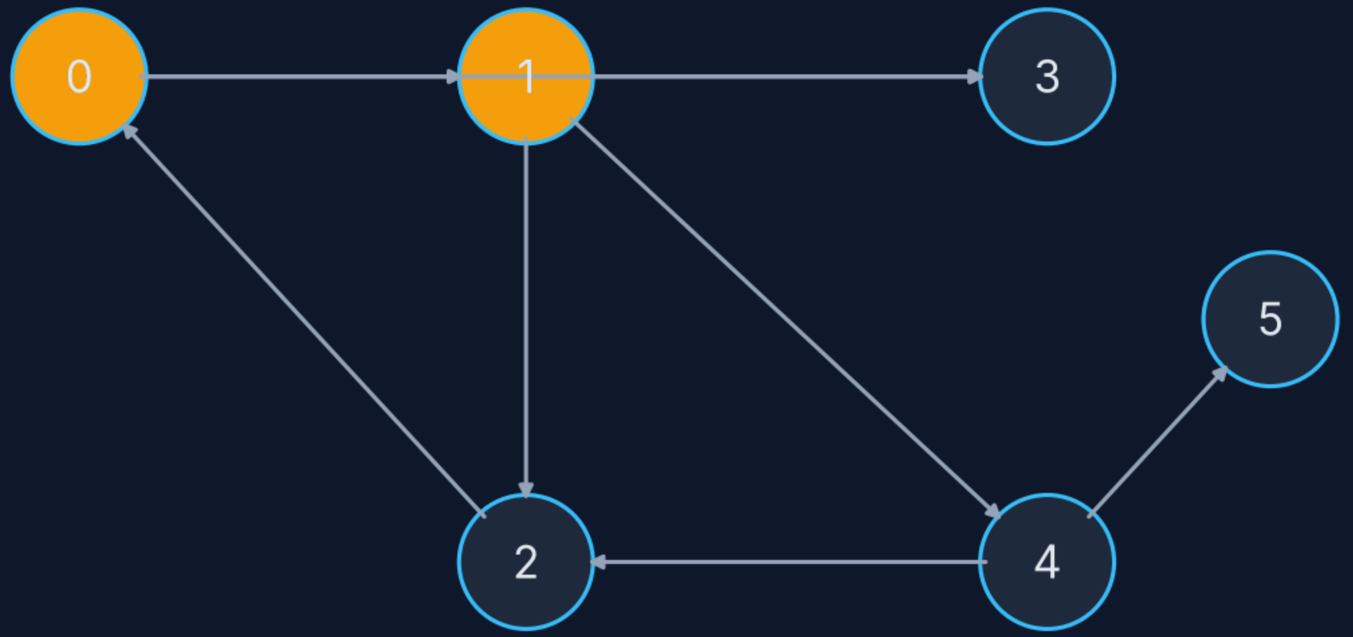

```
grey = on stack

dark = finished

colors: [1, 0, 0, 0, 0, 0]
```

Enter 0 (grey = on the recursion stack)

Worked example



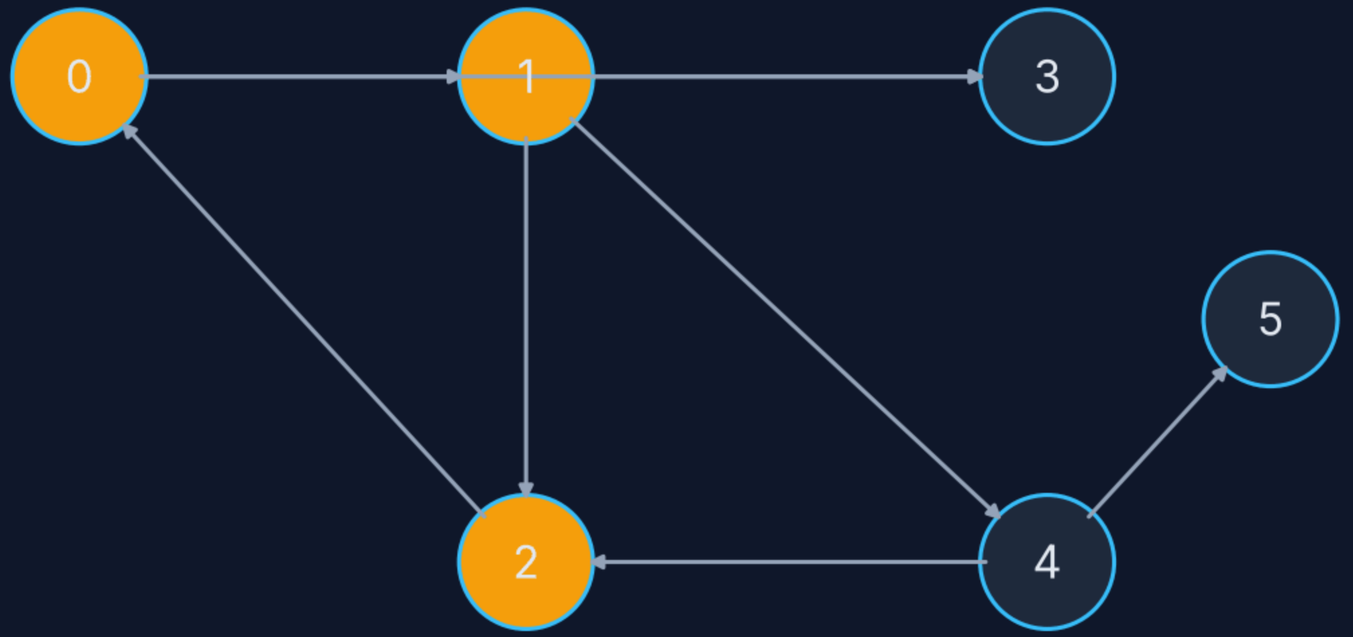
grey = on stack

dark = finished

colors: [1, 1, 0, 0, 0, 0]

Enter 1 (grey = on the recursion stack)

Worked example



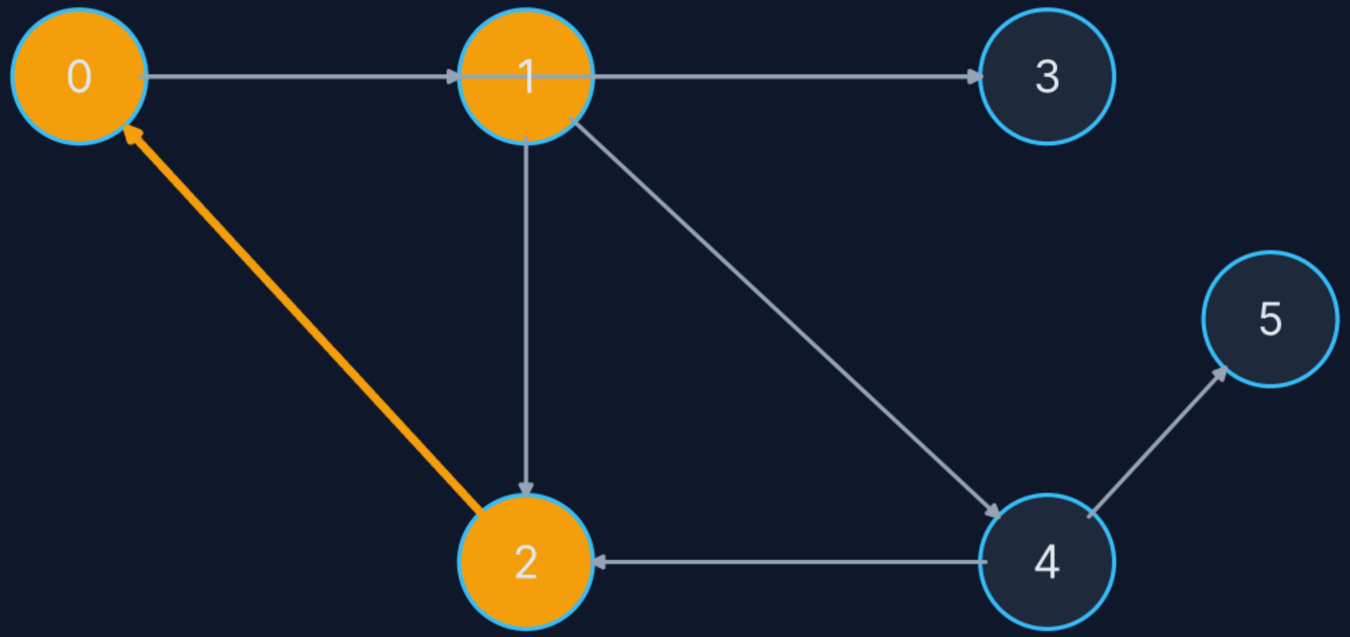
grey = on stack

dark = finished

colors: [1, 1, 1, 0, 0, 0]

Enter 2 (grey = on the recursion stack)

Worked example




```
edge 2 → 0

0 is grey → back edge

cycle!
```

Edge 2 → 0 hits a grey node → cycle

Java solution

```java
public class CycleInGraph {
    public static boolean cycleInGraph(int[][] edges) {
        int[] color = new int[edges.length];
        for (int v = 0; v < edges.length; v++) if (color[v] == 0 && dfs(v, edges, color)) return true;
        return false;
    }

    private static boolean dfs(int v, int[][] edges, int[] color) {
        color[v] = 1;
        for (int w : edges[v]) {
            if (color[w] == 1) return true;
            if (color[w] == 0 && dfs(w, edges, color)) return true;
        }
        color[v] = 2;
        return false;
    }
}
```

Complexity

- Time: O(v + e)
- Space: O(v)

Edge cases & pitfalls

- Visited ≠ on-stack: finished (black) nodes are safe
- Self-loops count as cycles
- Disconnected graphs: start DFS from every node

Interview tips & follow-ups

- Explain white/grey/black clearly
- Follow-up: undirected graphs (ignore the parent edge)

Recap

- A cycle exists if depth-first search meets a node that's still on the current recursion stack.
- Complexity: O(v + e)